- Build participants' skills in being self-directed learners. Teach goal setting approaches such as SMART. Reframe occasional failures as opportunities to learn what must be done to improve (Kivunja, 2015). If possible, offer assessment re-attempts when participants are ready to try again, even it's only for their own reflection.



Practitioner Competency Framework

- SFS 2.8.2 Encourages a growth mindset by praising and commenting on participants' efforts and processes rather than qualities or traits that reinforce a fixed mindset
- SFS 3.7.4 Provides constructive feedback to encourage adaptability
- SFS 3.7.6 Encourages participants to see errors and mistakes as growth opportunities rather than setbacks or failures

2 Focus and Emotional Regulation

Focus and Emotional Regulation have been a concern for many mentors, service providers and employers. Both personal and training stressors can affect participants to the point where they need immediate and longer-term strategies to deal with challenges. Participants with mental health challenges and test anxiety benefit from learning strategies to deal with stressors.

Instructional Strategies:

- Teach coping mechanisms that can calm stress and anxiety and help students develop adaptability skills to changing and stressful circumstances. Mindfulness-based interventions can reduce students' stress and anxiety, increase optimism and improve social and cognitive skills (Schonert-Reichl et al., 2015; see also Schonert-Reichl & Lawlor, 2010; Beauchemin et al., 2008; as cited in OECD, 2019).
- Emotional regulation is key to perseverance and adaptability. Teach acceptance-based and perspective-taking approaches to stressful situations (Wojnarowska et al., 2020).

Engagement

Strategically integrate social-emotional Skills for Success into training to maximize learning and skill gains.

Implementation Guidance (p. 111)



Practitioner Competency Framework

- SFS 2.8 Supports social-emotional well-being to develop self-regulation skills
- SFS 2.8.3 Supports participants with anxiety, trauma and other adverse experiences and behaviours
- SFS 2.8.4 Uses interventions to improve participants' ability to focus, minimize distractions and regulate emotions

3 Adaptability Training for Leaders and Work Contexts

Standard training practices emphasize expertise in typical tasks but that is not sufficient to support leaders as work grows in complexity and change. They need to develop skills in "adaptive expertise," the ability to deal with unanticipated, unfamiliar situations (Nelson et al., 2010).

Instructional Strategies:

- Incorporate variety into practice scenarios or training that requires participants to shift their existing strategies to adapt to new circumstances. The variations should help participants conceptualize how different problems require different solutions (Nelson et al., 2010).
- Provide strategic information and guidance to change perspective before, during and after scenarios. Give preparatory instructions to help participants understand relationships and goals for everyone involved, prime them to think about upcoming problems and to regulate emotions better. During and after the scenario, give feedback to help participants improve ways of thinking about/reframing a problem, and to improve future performance (Nelson et al., 2010).
- Support leaders with instruction in intersectionality, inclusion and diversity.

Under represented Groups

Engagement

Coordinate with sectoral organizations and employers to align training with skill needs in the workplace.

Implementation Guidance (p. 107)

Assessment Considerations

Assessment models for social-emotional skills, including Adaptability, are still under development. Given the multifaceted nature of adaptability, there may be need to develop or use multiple assessments to assess each sub-facet separately (SRDC, 2021). The *Research Report to Support the Launch of Skills for Success* (2021) outlines the pros and cons of several methods, such as situational judgement tests and subjective reports (see Table 2, p. 63). Assessments that are likely to be used in training programs will need to strike a balance between rigour and usability.

Suggested approaches:

- Develop a self-reporting rubric for participants, coordinating with the Skills for Success components of Adaptability. Self-assessment is an important component of efforts to develop thinking skills and self-concepts of adaptability.
- Develop a competency assessment such as The University of Alberta's *Competency Assessment for Demonstrating Adaptability and Flexibility.*

Proficiency Levels:

Proficiency levels are the level at which a person demonstrates a particular skill. These levels help practitioners build assessment tools to measure participant's skills. The Office of Skills for Success will refine the proficiency levels over time.

Preliminary Proficiency Levels

Entry: You can follow direction to adjust and complete plans, tasks, and goals. You can do this in response to expected and unexpected changes requiring minor adjustment or learning that is provided. You can stay positive, persist, and manage emotions in response to minor stress.

Intermediate: You can adjust and complete plans, tasks, and goals with some supervision. You can do this in response to expected and unexpected changes requiring moderate adjustment or learning with some resources provided. You can stay positive, persist, and manage emotions in response to moderate stress.

Advanced: You can adjust plans, tasks, and goals independently. You can do this in response to expected and unexpected complex changes requiring significant adjustment or learning that is self-directed using diverse resources. You can stay positive, persist, and manage emotions in response to high stress.

Research Report to Support the Launch of Skills for Success (p. 49)

RESOURCES:

- ABC Literacy. (2023). *Adaptability.* UP Skills for Work. https://upskillsforwork.ca/wp-content/uploads/ABC-UP-Adaptability.pdf
- Barclays. (2023). *Developing adaptability - adaptability lessons: LifeSkills.* Barclays Life Skills. https://barclayslifeskills.com/help-others/lessons/adaptability/
- FutureLearn. (2021). *How to build adaptability and Emotional Intelligence Skills.* https://www.futurelearn.com/info/blog/adaptability-emotional-intelligence-skills
- Skills/Compétences Canada. (2022). *Workbook: Adaptability. Skills for Success.* https://yourskillsforsuccess.com/workbook/adaptability-creativity/
- University of Alberta. (n.d). *Competency assessment for demonstrating adaptability and flexibility.* https://www.ualberta.ca/human-resources-health-safety-environment/media-library/learning-and-development/pathways-learning-series/adaptandflexassessmentguide2017.pdf



GRAB AND GO TRAINING IDEAS

Activity	Adaptability Components	Instructional Tips	Sector-Specific
At the start of a program or module, teach participants the SMART method to create clear, attainable, and meaningful goals for themselves. To ensure goals are clear and realistic, they should be: • **S**pecific • **M**easurable • **A**chievable • **R**elevant • **T**ime-bound	4. Set or adjust your goals and expectations • *Set goals and expectations based on your skill sets, available resources and supports* • *Define expectations and standards for reaching goals* 5. Plan and prioritize • *Define tasks, milestones, and longer-term strategies to achieve goals* • *Prioritize and choose an order of tasks according to your circumstances*	Ask participants to maintain a learning journal to track their goals and assessment results to keep perspective on effort and achievements. Schedule time for participants to reflect on their goals periodically during the program and adjust goals as their experience increases. Schedule time for a final reflection for their personal journey in continuous learning.	Contextualize goal-setting to the sector.  Guide new pastry chefs training to work in an upscale restaurant to develop goals to master new techniques. Offer support and strategies to help them meet their goals.
Integrate mindfulness training into programs. Teach participants to acknowledge feeling distracted or stressed, and how to recentre themselves using body scans and focus on their breathing.	1. Demonstrate responsibility • *Focus your attention on the current task* • *Minimize distractions* 3. Regulate your emotions when appropriate • *Stay calm when you are facing setbacks* • *Recover when you are not able to stay calm*	Incorporate a few minutes of quiet thought to start the class session, or to prepare for an assessment with positive self-talk. Provide recommendations for mobile apps for mindfulness breaks.	Contextualize emotional regulation skills to the sector.  Discuss potential jobsite stressors with apprentices before they leave training to enter the workforce. Discuss acceptance of the discomfort of being new on the job and being fully present in the learning situation.
Work with leaders to anticipate and develop a training model for workers types they haven't trained before, such as youth or workers with language needs or learning differences. What differences do they expect? What training strategies need to be adapted and how? How can you train, support and retain those workers?	2. Persist and persevere • *Anticipate changes* • *Reflect and evaluate what changes have happened and what is coming* • *If appropriate, modify your plans and approaches to reflect changing circumstances*	Encourage participants to brainstorm alternate points of view and challenge their own and their peers' assumptions. Build intercultural competence by discussing how learning needs may vary across worker demographics.	Contextualize adaptability in leadership training to the sector.  Work with management and human resources to determine trends in hiring for entry positions. Adapt or reconfigure training models to be more effective for the likely new hires.

Creativity & Innovation

Creativity and innovation is your ability to imagine, develop, express, encourage, and apply ideas in ways that are novel, unexpected, or challenge existing methods and norms.

Why are Creativity & Innovation Skills Important?

Creativity and innovation skills help you come up with new, unique, or "outside the box" ideas or to approach something completely differently than in the past. A curious mindset that finds inspiration from a broad range of experiences and perspectives helps develop creativity and innovation skills. Employers are increasingly seeking people who can apply creativity and innovation skills to their work in our increasingly diverse settings, and to come up with new solutions or approaches to tackling challenges. With strong creativity and innovation skills, you can also support and inspire others to develop their own creativity and innovation.

Creativity & Innovation Components

1. Use your imagination and curiosity
2. Identify opportunities for you to innovate
3. Generate ideas that are novel to yourself or others
4. Develop your ideas
5. Apply your ideas
6. Facilitate a creative and innovative environment for yourself and others

Research Report to Support the Launch of Skills for Success (p. 31)

Sector	Workplace Examples
Construction	Insulators decide how to cut materials accurately while also minimizing waste.
Manufacturing	Pulp Mill Machine Operators may deal with false readings on the computer. For example, readings may indicate that stock consistency is fine, but in fact it is too thick and won't go through the pumps and pipes.
Tourism	Cooks respond to market trends and seasonal ingredient availability to create new menu items. They may need to work with dietary limitations and food cost controls.
Forestry	Forestry Technicians work with local communities to determine appropriate measures to mitigate detrimental effects of collapsed riverbanks due to road-building operations, following rainy weather.

Transitioning from Essential Skills to Skills for Success

The Essential Skills Framework included elements of Creativity & Innovation woven into the skills of Communication, Thinking, Working with Others, and Continuous Learning. While the concept of Creativity & Innovation was not explicitly stated, it was integrated within activities involving creation, troubleshooting, adapting, dealing with challenging communication situations as well as learning new skills.

In the Skills for Success model, Creativity & Innovation responds to the growing importance of soft and social-emotional skills in the labour market. It encourages learners from diverse groups to apply skills learned in other contexts to the workplace. Creativity & Innovation may raise awareness and increase respect and valuing of the traditional knowledge and creative skills of Indigenous learners, which are not always formally recognized.

Creativity & Innovation is a skill that interacts and interconnects with other skills in real-world situations.

Evidence-Based Instructional Strategies

The following instructional strategies have been selected to highlight established or promising practices within the Skills for Success field and/or significant research that supports the instruction of creativity and innovation skills. Teaching for creativity and critical thinking can encompass a wide range of instruction methods. It also requires a redefinition of roles, particularly, moving away from the notion that instruction should lead participants' thinking towards a clear and pre-determined path (OECD, 2019).

1 Real World Applications

Training for creativity and innovation should be applied to realistic community, classroom and workplace contexts in order to be most effective. Adults learn more efficiently when they are striving to achieve some goal and can apply what they learned (Janiszewska-Szczepanik, 2020).

Instructional Strategies:

- Generating novel ideas can be the most difficult part for learners as many people think of creativity as a personality attribute, not a learnable skill, and do not see themselves as "creative types." Help participants get over this internal block by facilitating brainstorming sessions on topics that are relevant to the participants. Encourage both practical and silly ideas to boost imaginative thinking. This step can "oil the wheels" for developing ideas further.
- Coach learners to build their creativity skills systematically, moving up in complexity and adding each component of the Creativity & Innovation Skills for Success model. Although different aspects of creativity do not always happen in a definite order, the systematic approach can be an effective strategy to make skill gains accessible by diverse learners.

- Presentation of material in the teaching session should be followed by applied practice to the subject matter you taught. These exercises should be designed to provide students with practice in applying strategies and heuristics in a realistic and more complex context, perhaps aligned with employment and job performance. The applied practice can be a product of some sort (a paper, presentation, performance, model etc,) to make the learning visible and tangible (OECD, 2019).

Engagement

Coordinate with sectoral organizations and employers to align training with skill needs in the workplace.

Implementation Guidance (p. 107)

Note: See Approach #2 in Section 2 to learn more about contextualizing instruction across skills, sectors and audiences.



Practitioner Competency Framework

- SFS 2.2.2 Creates task-based learning activities based on real-world, workplace applications
- SFS 3.4.3 Integrates discussion, brainstorming and collaboration into creativity and innovation activities
- DELIVER 2.1.3 Incorporates real-world tasks and authentic workplace materials into learning activities, including culturally and contextually relevant materials

2 The Creative Problem-Solving (CPS) Model

Employers highly value their teams' ability to think outside the box with respect to troubleshooting and problem solving. Cognitive models such as the Osborn-Parnes Creative Problem Solving (CPS) model are particularly effective to build divergent thinking and problem solving, and also to generate creative products, attitudes and behavior (Scott et al., 2004).

Instructional Strategies:

- Model and provide explicit instruction in structured problem breakdown using checklists, feature comparisons and steps. Interestingly, training techniques that use such analytic processes are more effective than those that emphasize free exploration to tackle new and ambiguous problems (Scott et al., 2004).
- The Creative Problem Solving process can be applied to solve open-ended problems, in three stages: Understand the problem, generate ideas; and implement ideas. Try to expand on this model, implementing the components in the Skills for Success model (*Research Report*, 2021, p. 31). Start each stage with brainstorming to generate ideas, and then use an evaluation phase to filter those possibilities (Davis, 2006, as cited in Tsai, 2013).
- Implement the use of creative problem-solving models in the instruction of other Skills for Success. For example, thinking creatively can be modelled and implemented in numeracy instruction to break down and evaluate a "plan of attack" for a complex math application. Draw attention to and discuss when learners intuitively applied the model in their learning. Help learners make the connection between learning this abstract skill with their longer-term goals.

Engagement

Strategically integrate social-emotional Skills for Success into training to maximize learning and skill gains.

Implementation Guidance (p. 111)



Practitioner Competency Framework

- SFS 3.4.1 Uses conceptual model(s) to help participants work systematically with information in creativity-based tasks
- SFS 3.4.3 Integrates discussion, brainstorming and collaboration into creativity and innovation activities
- SFS 3.6.1 Provides explicit instruction on the basic elements and procedures of problem solving

3 Engage Underrepresented Groups

Practitioners have stated that the integration of social-emotional skills in the Skills for Success framework provides the opportunity to increase inclusivity by engaging underrepresented groups (Indigenous peoples, racialized people, newcomers, people with disabilities, and members of the LGTBQ2+ community) in training (*Implementation Guidance*, 2022, p.104).



Under represented Groups

Instructional Strategies:

- Create a learning environment where students feel safe to take risks in thinking and expressing their ideas. An open and non-judgmental classroom culture should have participants agreeing to rules for sharing feedback: criticism is not permitted, have respect for the individual and be open to novel ideas of yourself and others.
- Draw on the traditional knowledge and creative skills of Indigenous and newcomer participants. Showing respect to diverse perspectives and making learners feel acknowledged can help participants to challenge their self-perceived limitations and to address internal blocks to creativity (*Research Report*, 2021, pp. 33-34).
- Support multi-barrier youth with a coaching approach that involves self-reflection, encouragement and safety (*Research Report*, 2021, p 37). Provide opportunities to self-reflect on growth and goals.

Engagement



Create safe learning spaces by leveraging the social emotional Skills for Success to build learners' self-efficacy, autonomy, and confidence.

Implementation Guidance (p. 109)

Assessment Considerations

Assessment models for social-emotional skills, including Creativity & Innovation, are still under development. The *Research Report to Support the Launch of Skills for Success* (2021) outlines the pros and cons of several methods, such as task-based assessments, situational judgement tests, and subjective reports (see Table 2, p. 55). Assessments that are most likely to be used in training programs will need to strike a balance between rigour and usability.

Suggested approaches:

- Use scoring rubrics to assess performance-based tasks
  - Checklist ratings may allow for better standardization
  - Holistic ratings capture overall feeling and nuance
- Develop a self-reporting rubric for participants, coordinating with the instructional scoring rubric. Self-assessment is an important component of efforts to develop thinking skills and self-concepts of creative productivity.
- Aim to assess these creative performance measures as per the Torrance Test of Creative Thinking: fluency, flexibility, originality, and elaboration (Rose & Lin, 1984).

Proficiency Levels:

Proficiency levels are the level at which a person demonstrates a particular skill. These levels help practitioners build assessment tools to measure participant's skills. The Office of Skills for Success will refine the proficiency levels over time.

Preliminary Proficiency Levels

Entry: You can generate a limited number of novel ideas under guidance and support. You are open to applying new ideas, but are quick to revert to norms and habits when the new ideas fail or face uncertainties.

Intermediate: You can generate a larger number of novel ideas on your own. You acknowledge and work with uncertainties, accept failures, and learn from failures to improve your ideas. You are receptive to new ideas from others.

Advanced: You can generate a wider range of novel ideas, with diverse dimensions of originality. You evaluate limitations of novel ideas and find ways to improve them to minimize failures and uncertainties. You facilitate an environment for others to be creative and innovative.

Research Report to Support the Launch of Skills for Success (p. 49)

RESOURCES:

- ABC Literacy. (2023). *Creativity and Innovation.* UP Skills for Work. https://upskillsforwork.ca/wp-content/uploads/ABC-UP-Creativity-Innovation-web.pdf

- Chamorro-Premuzic, T. (2015, February 23). *You can teach someone to be more creative.* Harvard Business Review. https://hbr.org/2015/02/you-can-teach-someone-to-be-more-creative

- Chlup, D. (2019). Cultivating creativity in adult literacy education settings. *International Journal of Literacy, Language, and Numeracy,* Fall 2019. https://files.eric.ed.gov/fulltext/EJ1246114.pdf

- Skills/Compétences Canada. (2022). *Workbook: Creativity and Innovation*. Skills for Success. https://yourskillsforsuccess.com/workbook/creativity-and-innovation/

- Vincent-Lancrin, S. et al. (2019). *Fostering students' creativity and critical thinking: What it means in school*. Educational Research and Innovation, OECD Publishing. https://read.oecd-ilibrary.org/education/fostering-students-creativity-and-critical-thinking_62212c37-en#page1



GRAB AND GO TRAINING IDEAS

Activity	Creativity Components	Instructional Tips	Sector-Specific
Teach and practice brainstorming. Apply the lesson to an activity that participants have existing experience with. Use lists, mind maps, sketches etc to demonstrate different ways participants can record their own idea generation. Debrief on connections and themes.	3. Generate ideas that are novel to yourself or others • *Deviate from existing processes, thinking and approaches* • *Use an inquisitive approach (e.g., ask questions even when there is no obvious answer)* • *Seek patterns where patterns may not be readily apparent* 4. Develop your ideas • *Expand on ideas and approaches*	Set the rules for brainstorming: No criticism, work for quantity of ideas, build on others' ideas, free thinking is encouraged. Focus on positive feedback to make it safe for participants to push past internal blocks. To encourage novel ideas, what would different characters say (a child, a superhero, a celebrity, a person with different abilities)?	Contextualize topics to the sector.  Carpentry students brainstorm different styles of 5-foot wooden fences that can be custom-built for a standard residential lot.
As a group, brainstorm workplace problems that require creative problem solving as opposed to established procedures. Choose one or two to apply a problem-solving model (CPS, 5 Whys, Lean Six Sigma). Would a diagram such as a mindmap or cause & effect (tree) diagram help organize and elicit creative responses and relationships?	1. Use your imagination and curiosity • *Imagine different situations and possibilities* • *Be open to new ideas without judging and setting limitations* 2. identify opportunities for you to innovate • *Challenge norms, habits, and preconceptions where appropriate*	Provide explicit instruction on the rules of brainstorming (as above) and on the structure of the organizational diagram. Provide opportunities for participants to "think outside the box". Provide templates for checklists, mindmaps and tree diagrams.	Contextualize creativity skills to the sector.  Discuss strategies to increase purchases of a foodie walking tour by implementing social media promotion.
Celebrate creativity and innovation demonstrated by participants as it is interwoven in learning any skills. What did you and participants see as growth in approaching learning or in demonstrating competence? How did participants use strategies and heuristics to deal with complex content? Participants can build a portfolio showing their growth and growth mindset.	5. Apply your ideas • *Expect failures* • *Learn from failures to improve* 6. Facilitate a creative and innovative environment for yourself and others • *Encourage habits and behaviors that facilitate creativity and innovation in yourself and others*	Post examples of creative work done by participants, with their permission. Use self-assessment tools to guide reflection and self-assessment.	Contextualize recognition and self-assessment of growth in creativity.  Submit summaries to local newspapers of experimental work your participants have done to reduce water demands in growing vegetable crops, that home gardeners can utilize.

Section 4

Appendix

This section offers resources to enhance your understanding of concepts presented in the Skills for Success Get Started Guide and facilitate your professional development. It includes three sample lesson plans* contextualized to specific sectors that demonstrate the best practices highlighted in the guide. Additionally, you can access a list of professional development workshops and courses for Skills for Success practitioners, as well as a bibliography of all the references and resources mentioned in the guide.

*More lesson plans coming soon!



Lesson Plan #1:
Reading an Imperial Measuring Tape (Numeracy Focus)



This lesson is intended primarily for the construction sector. An instructor can use this lesson as an experiential learning, hands-on activity to apply learning about fractions and the imperial measuring system.

Time: 1.5 hours

Skill Components

This lesson focuses primarily on Numeracy Components #2 and #4.

Numeracy Components

2. Identify the mathematical information
   - Locate key details, concepts and other mathematical information
4. Apply mathematical operations and tools you will need to answer the question
   - Measure

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Collaboration Components

1. Work well with other people
   - Engage in trust building behaviours
   - Adhere to social and organizational rules
   - Assess strengths and weaknesses of yourself and others
5. Achieve a common goal with others
   - Take responsibility to make contributions and complete tasks
   - Ensure opportunities for others to contribute

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Materials & Resources

Handout

- List of items to measure precisely. Specify the dimensions (length, diameter, thickness, etc.)
- See Handout 1: Measuring in Imperial

Materials

- Standard school imperial rulers, one per participant. Marked in inches and 16ths of an inch.
- Standard Imperial measuring tapes, marked with numbers for whole inches and lines indicating 16ths of an inch. One measuring tape per pair of participants.
- Pencils or pens.
- Trade-specific items. Suggestions: pieces of dimensional lumber, screws, framing hardware, sections of ABS or metal piping, steel structural members, wrenches, conduit, cable, stakes. Include a variety of sizes and weights of materials.

Pre-requisite Knowledge & Skills

Participants need to have learned math foundational skills for working with fractions: equivalent fractions, simplifying fractions. They should have basic skills in reading a small imperial ruler marked in inches and lines indicating fractions of an inch, be able to identify markings for fractional inches, and state fractions of an inch in lowest terms.

The complexity of the numeracy and collaboration tasks can be adapted for a variety of levels.

Virtual Delivery Tips

This lesson can be delivered in-person or virtually. Here are a few ideas to adapt the lesson for a virtual audience:

- Available Materials: Ensure participants have access to measurement tools such as rulers and measuring tapes at their virtual locations. Adapt measurement tasks and handouts to reflect objects commonly available in learner homes.
- Digital Handouts: Convert the handouts into digital formats that can be easily distributed to participants prior to the session.
- Virtual Whiteboard: Utilize a virtual whiteboard or annotation tool to demonstrate how to read whole inches and fractional markings.
- Interactive Polls or Quizzes: Incorporate interactive polls or quizzes to gauge participants' understanding of reading measurements.
- Breakout Rooms: Set up virtual breakout rooms for participants to work together in pairs or small groups.
- Screen Sharing: Encourage participants to use screen sharing to demonstrate their measurement techniques or share their completed handouts with the larger group during the debriefing session.

LESSON ACTIVITIES	
1	*Opening discussion* Ask participants what their experience has been with using measuring tapes in work or home contexts. Did they use imperial or metric measuring tapes?
2	*Activate background knowledge* With participants reading standard school rulers, review how to read whole inches and line markings for fractional inches. Remind participants to state fractional inches in lowest terms.
3	*Handout 1* Provide participants with handouts of the measurements required to be made in and around the classroom. Set up pairs or small groups of participants to work together. Distribute one measuring tape per pair. This is a good activity for peer support as those with some experience can help others. You can be strategic in pairing up participants. This activity is often very successful when run as experiential learning and helps build a collaborative learning culture in the class. Don't pre-teach strategies but let the participants experiment with what to do for irregular items. Customize your handout to the learning space and potentially outdoors or the shop. Start with regular, rectangular rigid shapes such as desks and doors. Increase the complexity by including irregular items that require creative thinking such as: • Flexible items like participants' heights • Over-height items such as the room height • Over-length items such as the hallway, which exceed the length of the tape measure • Round items, where the largest measurement is the diameter • Inside dimensions such as inside windowsills • Small items that are more easily measured starting at the 1-inch mark

LESSON ACTIVITIES	
4	*Discuss* Discuss the assignment and expectations for respectful behaviour in the classroom and surrounding area so as not to disrupt other groups. Discuss behaviours for effective collaboration amongst team members.
5	*Practice* Give groups a time limit of 20 – 30 minutes to take measurements, record them on the handout and bring back the information to the classroom to debrief. Observe participants' collaborative behaviours, such as the different roles that team members take on, their contributions to the task and ability to work together. Make note of how participants apply creativity to problem solve – do they brainstorm different options, expand on each other's ideas, and learn from failures to improve?
6	*Discuss* Collect the measurements for each group without judgement of right or wrong. Ask participants to describe what strategies they used for irregular items and dealing with the tool. Encourage a growth mindset by praising and commenting on participants efforts and creativity in problem-solving. Debrief the collaboration and creativity skills required, asking participants to identify their strengths and areas for growth as individuals and as a team.
7	*Discuss* Extend learning by asking participants to convert some of the measurements that were taken in inches (like their heights) into feet and inches (eg. 64 in = 5 ft and 4 in). Extend learning by asking participants to calculate the difference between two measurements, such as the tallest participant and the instructor, or total distance between two adjacent points.

Follow-up

Follow up with YouTube videos of carpenters' tips on tape measure use.

Conduct a contest to see who can extend their tape measure the furthest before it falls.

Next lesson, ask participants if they can recall or estimate some of the dimensions they measured. Using known dimensions, such as their height or the length of a desk, they can estimate longer distances. That can be very helpful in the field.

The next lesson can apply skills in reading a measurement on a graphic of a tape measure. Participants are asked to add on or subtract inches and fractions of an inch. For example, given a mark at 64 ½ inches, participants are asked to and add on 3 ¾ inch. What is the total measurement?



Handout 1

Measuring in Imperial

Measure the following items in inches.
Measure accurately to the nearest 16th of an inch.

1) Baseboard heater in classroom (length)

2) Classroom door (length, width, thickness)



3) Diagonal length of the TV monitor

4) Perimeter (total distance around the outside) of cement planter at the building entrance

5) Blue pen (length and diameter)



6) Diameter of a lunchroom table (at the widest point)

7) Circumference of (distance around) column in classroom



8) Your heights

9) Your arm span from finger tip to finger tip

10) Width of a regular parking stall in the parking lot, centre-to-centre of the line markings

11) Height of the classroom

12) Length of the hallway at the longest point

Lesson Plan #2:
Root Cause Analysis (Problem Solving Focus)

This root cause analysis lesson for production workers or supervisors in the manufacturing sector is part of a series of lessons where participants will be working in teams to complete a problem-solving project in their workplace.

Time: 1.5 hours

Skill Components

This lesson focuses primarily on Problem Solving Component #3.

Materials & Resources

Handout

- Handouts: 1. Fishbone Diagram Template,
  2. Five Whys Template

Problem Solving Components

3. Analyze the issue
   - Think critically about the issue using the information gathered
   - Break down the issue into smaller parts
   - Seek patterns, make connections across information
   - Identify possible cause-and-effect linkages

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Collaboration Components

2. Value diversity and inclusivity of others
   - Respond without judging people for their different opinions, ideas, and views

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Pre-requisite Knowledge & Skills

At this stage in the problem-solving process, participants will have identified the issue to be addressed and gathered information about the problem. The instructor will have taught brainstorming rules and techniques in a previous lesson, so participants would understand that free thinking is encouraged and criticism should be avoided.

The complexity of the problem-solving and collaboration tasks can be adapted for a variety of levels.

Virtual Delivery Tips

This lesson can be delivered in-person or virtually. Here are a few ideas to adapt the lesson for a virtual audience:

- Available Materials: Ensure participants have access to the necessary handouts digitally. Share the Fishbone Diagram Template and Five Whys Template electronically before the session.
- Virtual Discussions: Use the virtual platform to facilitate opening discussions and activate background knowledge. Ask participants to share their experiences with root cause analysis tools and techniques, utilizing features like chat or raised hands for participation.
- Visual Tools: Utilize screen sharing to present the structure and purpose of the fishbone diagram. Guide participants through the process of identifying categories and demonstrate how to use the diagram effectively.
- Breakout Rooms: Organize virtual breakout rooms for participant teams to collaborate on completing their fishbone diagrams.
- Interactive Activities: Incorporate interactive activities using virtual tools for brainstorming and completing the 5 Whys exercise. Encourage participants to share their screens and collaborate in real-time.
- Virtual Presentations: Allow participant teams to present their fishbone diagrams and 5 Whys analyses to the class using screen sharing. Allocate time for each team to explain their process and discuss their findings.

LESSON ACTIVITIES	
1	*Opening discussion* Ask participant teams to present the information they have collected about their problem since the previous session. This will include the data they feel quantifies the impact of the problem as well as the context they have gathered by speaking with stakeholders that are affected by the problem. Explain that today's session will be about analyzing this data to identify the root cause of the issue.
2	*Activate background knowledge* Ask participants to share their understanding of and experience with root cause analysis. What tools do they know about and what tools do they currently use?

LESSON ACTIVITIES	
3	*Handout 1* Instructor introduces the structure and purpose of the fishbone diagram. A fishbone diagram is a visual tool used in root cause analysis to identify and categorize potential causes of a problem, facilitating structured problem-solving by breaking down complex issues into manageable components. Help participants to identify categories that can be used as headings: equipment, process, people, materials, environment, management, etc. or leave categories blank for participants to decide. Present a fishbone diagram on a similar problem to demonstrate how it can be used. Remind participants of the brainstorming and collaborative strategies that have been previously taught.
4	*Practice* Participant teams brainstorm to complete their fishbone diagram using the data and context they have collected since the previous class. Participant teams then present their fishbone to the class. Ask participants to explain their process and any successes or pit-falls they experienced along the way.
5	*Handout 2* Introduce the 5 Whys tool, which involves asking "why" multiple times (typically five) to dig deeper into the causes of a problem and uncover its underlying root cause, helping to address issues at their source. Explain that it can be used to analyze the problem as a whole or one of the causes that was identified through completing the fishbone diagram.
6	*Practice* Participant teams complete the 5 Whys activity and present to the class.

LESSON ACTIVITIES	
7	*Summary* Debrief the activities. Ask participant teams to summarize the root causes they have identified through using the two tools. Which tool was more effective? Was it helpful to look at the issue from different angles? Ask participants to discuss the collaboration skills they used and to identify their strengths and areas for growth as individuals and as a team.

Follow-up

Instructor explains that in the next class they will be using the root causes they identified to develop routes of action to solve the problem, emphasizing that in depth information gathering and root cause analysis ensures that solutions fully address the problem and not just the symptoms.